Stability and Performance Evaluation of Wireless Tele-Control System for MIMO Plant

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***Abstract*— In this research, we aim to evaluate the performance and stability of a jointed MIMO (Multi Input and Multi Output) plant and wireless communication system. In order to realize a distant control wireless communication can be applied to control system. The advantage of Wireless Tele-Control system is to minimize the weight of control object and can also be applied on unmanned vehicle system. However, In Wireless communication system channels are occurred due to utilization of wireless networks that can cause multipath channel. A multipath channel consists of accumulated delayed and attenuated reference signal. In wireless Tele-Control system, output signal are fed back to the controller in order to stabilize the closed-loop or compensate the performances. Due to this matter we have two multipath channels in the closed-loop system. One is feedforward channel and the other is feedback channel. Because of existence of multipath channel, control input and feedback signal are affected by multipath channels. As a result undesired signal are observed due to existence of multipath channel and may cause the closed-loop instability or performance degradation. General Speaking, design of controller with multipath channel becomes stiff problem to satisfy the stability of the closed-loop. Thus, in order to simplify the design of controller, we have jointed equalizers in the control loop. Meanwhile the effects of channels are reduced by equalizer which is consisted of FIR (Finite Impulse Response) filter. The stability and performance of the closed-loop system can be evaluated by step response. The plant is set to be a drone that aims to control the attitude. In conclusion we discussed about the performance and stability of Wireless Tele-Control system in frequency domain. Eventually we could confirm the simplification of controller design for MIMO and Wireless Tele-Control System.**

Keywords: Wireless Tele-Control System, Multipath Channel, Equalization, MIMO system, Unmanned Aerial Vehicles

1 Introduction

The utilization of Wireless Tele-Control system is one of the significant issues in the servo systems. Especially, when system requires control in distant. The advantage of Wireless Tele-Control system is that maintenance and management of controller can be done easily since controller is located in observation center and plant may be located in distant. One more thing is that by Wireless Tele-Control system since controller is not loaded on plant, it could be considered as reduction of load in plant and makes system performance enhanced. Let us clarify the Wireless Tele-Control system. Basically, in Wireless Tele-Control system they are always two channels. One is the feedforward channel to send the optimal or compensated input to the control plant and the other one is the feedback channel since output signal should be sent to the controller side in order to calculate the error and to minimize it or stabilize the closed-loop. So, these channels are disadvantages of utilization of Wireless Tele-Control system. First of all due to the usage of communication system in the closed-loop system we would have some impairment such as phase noise, Doppler effects, frequency offset, delays and attenuations. The mentioned impairment can be solved by implanting the system that has high function capabilities. Therefore, phase noise, Doppler effects, frequency offset can be repaired by installing the advanced function capability. However, the received signal should be equalized to get the original information from sender. Therefore, in order to get the exact data from sender it is required to equalize the received signal. The received signal may be distracted by the multipath channel. Multipath channel effect occurs concerning the circumstances of the environment of control plant. In other words, multipath channel is inclusion of accumulated delayed and attenuated direct path signal. Even though sender has sent the original signal but interfered signal will be received in receiver side. Thus, equalization of signal is required in receiver side. For equalization, first we have to compose the replica of the unknown channel. The composition of the replica of the unknown channel can be done by FIR adaptive filter. However, the composition of the replica channel is not sufficient. In order to get reference signal it is required to realize the inverse transfer function of replica Channel. Therefore, the inverse channel is realized after the receiving the distracted signal. This has role of equalizing the received signal. These processes should be implemented in two different stages. One is the feedforward side of the receiver and the other one is the feedback part of the receiver since we have round trip multipath channel in the closed-loop system. After realizing the equalizer, implementation can be done in the closed-loop system. Furthermore, controller can be designed according to plant without considering multipath channel. After designing a controller equalizers and controller are jointed in cascade. Thus, controller and equalizers are jointed in the closed loop system. In Next chapter we will introduce unmanned aerial vehicles and controller design of it.

2 Design of Feedback Controller for Unmanned Aerial Vehicles

In this chapter we introduce briefly design of an unstable system in order to stabilize the closed loop system. The plant is set to be unmanned aerial vehicles. First of all let us introduce unmanned aerial vehicles briefly.

The applications of unmanned aerial vehicles have been extended in both military and civilian fields around the world in the recent years. Unmanned aerial vehicles are used in all military ranging from investigation, monitoring, intelligence gathering, battlefield, distant investigation and several other aims. As well Civilian applications include remote sensing, transport, exploration, and scientific research. Because of vast application unmanned aerial vehicles is expected to be in vogue. Therefore, in the view of reliability of stabilization and performance, even in drastically changing environment such as strong storm unmanned aerial vehicles should be operated to do its duty. Therefore, controller required to be installed in the closed loop system. However, as we have mentioned the mass of unmanned aerial vehicles is not recommended to be increased as a stability and performance point of view since it may cause instability and performance degradation. Thus Wireless Tele-Control system is proposed. Following shows how to stabilize a plant.

First of all let us consider a plant which is that $G = \frac{1}{s^2}$ for $a > 0$. A basic PID (proportion integral and derivative) feedback controller can be considered as

$$K = Kp\left(1 + \frac{1}{T_i s} + \frac{T_d s}{as+1}\right) \quad (1)$$

that set (K_p, T_i, T_d, a) are tuned to make the closed loop system internally stable.

Here, K_p is proportional gain, T_i is integral gain, T_d is derivative gain and a is derivative approximated gain. The following conditions should be satisfied to maintain internal stability and performance enhancement of the closed loop system.

(1)- If and only if the real part of solutions of characteristic equation are less than zero.

Characteristic equation becomes as follows:

$$s^4 + \frac{1}{a}s^3 + K_p\left(1 + \frac{T_d}{a}\right)s^2 + K_p\left(\frac{1}{T_i} + \frac{1}{a}\right)s + \frac{K_p}{aT_i} = 0 \quad .$$

If the above equation's pole is set of number such as

$$\Pi = \left(p_i, \forall i = 1,2,3,4 \mid p_i \in C\right) \quad (2)$$

Stability condition is Re (Π) < 0.

(2)- Frequency response of Sensitivity function *S(jω)* should contain the following specification.

ω_b : Band width frequency

$$For\ \omega < \omega_b \quad | S(j\omega) | << 0\ [dB]$$

$$For\ \omega > \omega_b \quad | S(j\omega) | \simeq 0\ [dB]$$

The above condition described that when for sensitivity in the low frequencies it has small gain most of interferences and disturbances with direct current component charactresitc that can affect as external disturbances are not influenced the closed loop system. For high frequency domain it is desirable to maintain 0 [dB] to reduce the tracking error.

(3)- Frequency response of Complementary sensitivity function *T(jω)* should contain following specification.

$$For\ \omega < \omega_b \quad | T(j\omega) | \simeq 0\ [dB]$$

$$For\ \omega > \omega_b \quad | T(j\omega) | << 0\ [dB]$$

The above condition described that when for Complementary sensitivity in the low frequencies it nearly equal to 0 [dB] that means for reference signal that contains direct current componenet, it becomes almost same in the output of plant that is desiable. For high frequencies, it is desirable drop to small gain since can reduce the affect of feedback noise and plant uncertainty.

(4)- Frequency response of open loop function $G(j\omega)K(j\omega)$ should contain following specification.

ω_c: Cross frequency

$$\text{For } \omega < \omega_c \quad |G(j\omega)K(j\omega)| >> 0 \text{ [dB]}$$
$$\text{For } \omega > \omega_c \quad |G(j\omega)K(j\omega)| << 0 \text{ [dB]}$$

The above condition describes that when open loop in low frequency domain contain large gain, it can enhance the performances and when it is in high ferqnecies gain should be small values to reduce the effect of uncertainty in plant and external disturbances.

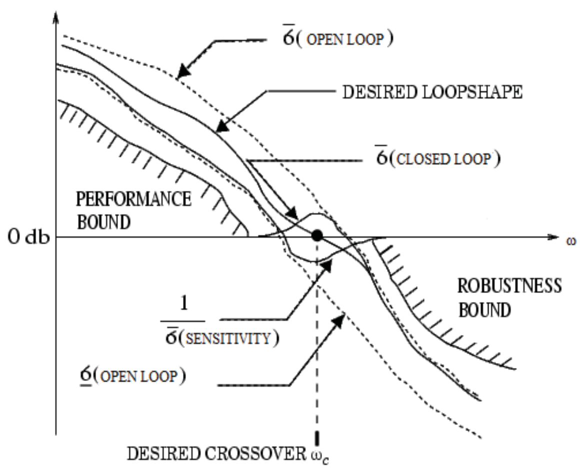

Fig.1 Singular value specifications on open loop, sensitivity, and closed loop

Here is an example of the described conditions.

$$plant: G(s) = \frac{1}{s^2} I_{3\times3}$$

$$Controller: K = Kp\left(1 + \frac{1}{T_i s} + \frac{T_d s}{as+1}\right) I_{3\times3}$$

$$Open \quad Loop: G_O(s) = G(s)K(s)$$

$$Sensitivity: S(s) = \left(I + G_O(s)\right)^{-1}$$

$$Complementary \quad Sensitivity: T(s) = G_O(s)\left(1 + G_O(s)\right)^{-1}$$

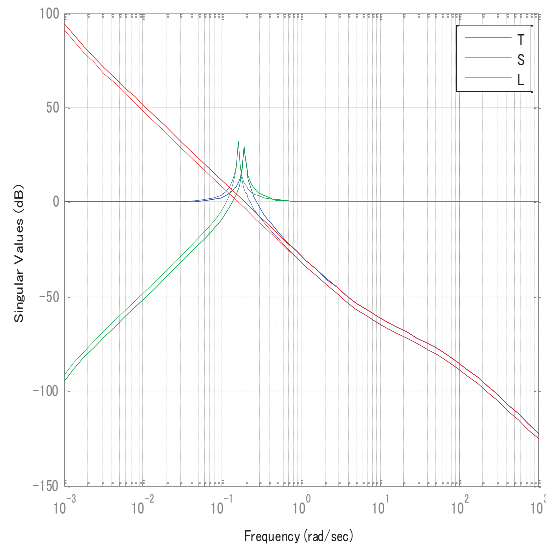

Fig.2 Specification of desired frequency response

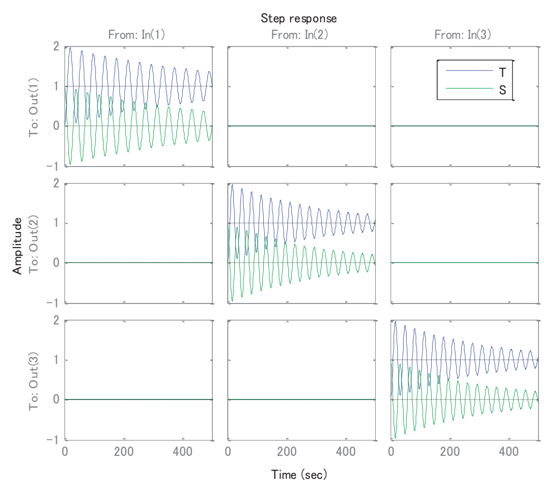

Fig.3 Specification of desired step response

As there are shown in the above figures, Fig.2 shows the desired frequency response of sensitivity, complementary sensitivity function and open loop, respectively. It is obvious that when frequency increase open loop and complementary sensitivity function overlapping each other. This matter can be confirmed mathematically.

As it shown on above equation, open loop has significant influence on the closed loop stability and performances. Therefore, controller design should be done by acquiring the plant characteristic and frequency response. However, in actual and practical cases, there would be parameters perturbation in plant and uncertainty always presents. Thus, a control scheme should overcome the uncertainty problem when design a controller. Next chapter joint system of feedforward and feedback control is introduced

3 Wireless Tele-Control System

So far we have discussed about the stability condition and performances of the closed loop system for ordinary case(without Channel). Here, let us define Wireless Tele-Control System as follows. Following figure shows the structure of Tele-Control system.

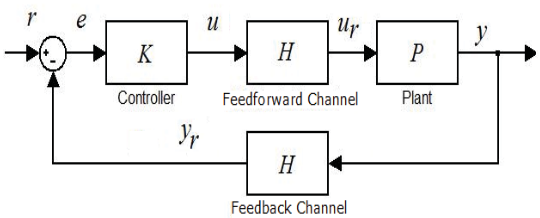

Fig. 4 Tele-Control System

Here H is Multipath channel and u_r, y_r are received input and received output signal, respectively. Through Fig.4, we can get the closed-loop system's transfer function according to following equations.

$$y = PHKe \tag{3}$$

$$e = r - Hy_r \tag{4}$$

Afterward we get the transfer function between r and y which is complementary sensitivity transfer function as follows.

$$y = \Delta_H PHKr \tag{5}$$

Where, $\Delta_H = (1 + HPHK)^{-1}$ stands for sensitivity transfer function which is from r to e.

As we can see in sensitivity function of the closed-loop system, it has been involved with Channel's square.
The stability condition of closed-loop system is

$$\bar{\sigma}\left((1+H(j\omega)P(j\omega)H(j\omega)K(j\omega))^{-1}P(j\omega)H(j\omega)K(j\omega)\right) < 1 \quad \forall\omega.$$

Which $\bar{\sigma}(.)$ indicate the maximum singular values.
However, due to existence of channels, it is stiff problem to satisfy the above condition. Therefore, our proposed method is to reduce the effect of the channel in the sensitivity function. The proposed method has shown in following Fig. 5.

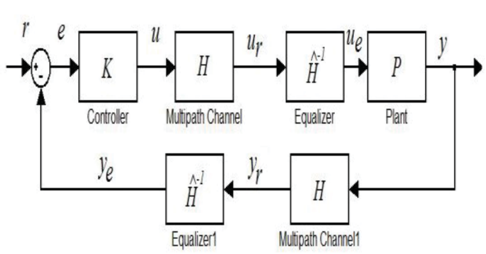

Fig. 5 Configuration of the proposed method

Here, y_e, u_e and $\hat{H}^{-1}$ stand for the equalized input signal, equalized output signal and Equalizer, respectively. $\hat{H}$ it self is the replica channel of H that is estimated with adaptive filter. However, before getting starting the proposed method let us see how we can design a controller for Tele-control system without considering channel equalizer. For fig.5, we can design a controller according to the plant only and neglect channels.
Assuming $H=I$ Then our complementary sensitivity function becomes

$$T(s) = (1 + P(s)K(s))^{-1} P(s)K(s)$$

and the necessity of stabilizing a MIMO system is that if and only if the complementary sensitivity function's maximum singular values is satisfying the equation (8).

For $s=j\omega$

$$\bar{\sigma}\left((1+P(j\omega)K(j\omega))^{-1}P(j\omega)K(j\omega)\right) < 1 \tag{8}$$

3 Reduction of Multipath Channel Effects in the Closed-loop System

As we have discussed previously, existence of the multipath channel make the systemunstable and it is very hard to determine the PID parameter that satisfies the small gain theorem which has been mentioned in Equation (7). Therefore, somehow the multipath channel should be eliminated in order to get rid of the instability. Thus, equalizer is required in the receiver side of the plant for the feedforward multipath channel and another equalizer is required in the controller side for feedback multipath channel. By implementation of the equalizer we can reduce the effect of the multipath channel. However, before equalizing the received signal estimation of multipath channel is required. Estimation of multipath channel can be done by adaptive filter. After reconstructing the replica of multipath channel, inversion of the replica channel should be implemented in cascade to vanish the multipath channel. In following figure the process of the proposed system is indicated in detail.

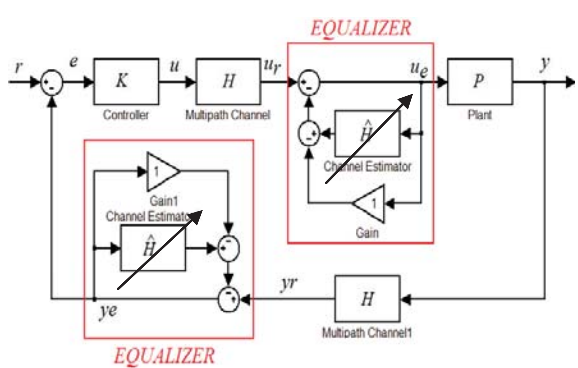

Fig. 6. Configuration of the proposed method in detail

$$y = KH\hat{H}^{-1}Pe \tag{9}$$

$$e = r - H\hat{H}^{-1}y_e \tag{10}$$

Afterward, we have as follows.

$$y = \left(1 + PK\left(H\hat{H}^{-1}\right)^2\right)^{-1} PH\hat{H}^{-1}Kr \tag{11}$$

That $\left(1 + PK\left(H\hat{H}^{-1}\right)^2\right)^{-1}$ is the sensitivity function of the proposed method.

If and only if $\hat{H} = H$, then we obtain the conventional feedback control system. In other words $\hat{H}^{-1}H = I$ that $\hat{H}^{-1}$ is an unitary matrix. However, the result of multiplication of $\hat{H}^{-1}H$ is rarely becomes identity matrix since replica channel cannot realize the exact characteristic of multipath channel. Nevertheless, we can reduce the effect of multipath channel in the sensitivity function and in the open loop.

4 Multipath Channels

Basically, multipath channel is consequences of the reflected desired signal or in other words accumulation of several attenuated and delayed reference signal. Especially, this phenomenon would be occurred easily and frequently in metropolitan ambit which comprised of high density of building and so. Also, it would occur in mountainous area as well. Following shows the signal composition in time domain of reflected signal which comprise of multipath channel.

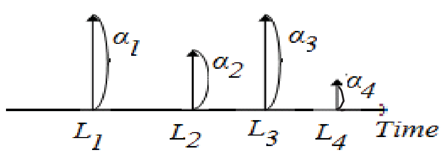

Fig. 7. Signal compositions in time domain

The mathematical model for a multipath channel H can be expressed as follows.

$$h(n) = \sum_i \alpha_i \delta(n - \tau_i) \tag{12}$$

Where, α and τ stands forattenuation andtime delay factor of multipath channel, respectively. As it is clear in equation (11), we need to estimate the multipath channel in order to get rid of instability in the closed-loop system. Therefore, for estimation of multipath channel FIR (finite impulse response) adaptive filter is utilized. The tap number of FIR adaptive filter concerns the length of multipath channel. Hence, the length of filter should exceed the length of multipath channel. Otherwise reconstruction of replica multipath channel becomes hard. In next chapter several adaptive filter algorithms are introduced.

6 Adaptive Filters

An adaptive algorithm [6-7]is a set of recursive equations used to adjust the weight vector of replica multipath channel H automatically to minimize the effect of multipath channel in sensitivity function. Such that the weight vector converges iteratively to the optimum solution that corresponds to the bottom of the performance surface, i.e. the minimum of MSE (Mean Square Error). The Least- Mean- Square (LMS) algorithm is the most widley used among various adaptive algorithm because of its using the negative gradient of the instantaneous squared error. In general expression for H that intent to adapt itself to H. The derivation of updated weight vector of LMS algorithm can be shown as follows. Here the adaptation done in time domain so we consider the *h(n)* as the imverse Laplace trasfer of *H(s)*. As well for $\hat{h}(n)$ is the inverse Laplace transfer of $\hat{H}$ (s). For feedforward equalizer let us define the error signal in the adaptive filter e_f(n).

$$e_f(n) = h(n) * u(n) - \hat{h}(n) * u_e(n) \tag{13}$$

According to stochastic gradient algorithm, we would have as follows. Here n and i are iteration and filter's tap number, respectively.

$$\hat{h}_i(n+1) = \hat{h}_i(n) - \mu \nabla e_f^2(n)$$

$$= \hat{h}_i(n+1) = \hat{h}_i(n) - \mu \frac{\partial e_f^2(n)}{\partial \hat{h}}$$

Calculation of gradient of square error is given by below.

$$\nabla e_f^2(n) = \frac{\partial}{\partial \hat{h}(n)} (h(n) - \hat{h}(n))(h(n) - \hat{h}(n))^T * (u(n) \times u^T(n))$$

$$= \frac{\partial}{\partial \hat{h}(n)} (h(n) - \hat{h}(n)) \times (h^T(n) - \hat{h}^T(n)) * (u(n) \times u_e^T(n))$$

$$= \frac{\partial}{\partial \hat{h}(n)} h(n) \times \hat{h}^T(n) - 2\frac{\partial}{\partial \hat{h}(n)} h(n) \times \hat{h}^T(n) + \frac{\partial}{\partial \hat{h}(n)} \hat{h}(n) \times \hat{h}^T(n)$$

$$= (0 - 2h(n) + 2\hat{h}(n)) * (u_e(n) \times u_e^T(n))$$

$$= -2(h(n) - \hat{h}(n)) * u(n) \times u_e^T(n)$$

$$= -2e_f(n)u_e^T(n)$$

Eventually, we obtain the following equation.

$$\hat{h}_i(n+1) = \hat{h}_i(n) - 2\mu e_f(n)u(n-i) \qquad (14)$$

where μ is the step size or convergence factor that determines the stability and the convergence rate ofthe algorithm.

In the case of Normalized LMS, the LMS algorithm normalizes the step size with respect to the input signal power.

$$\hat{h}_i(n+1) = \hat{h}_i(n) - \frac{2\mu e_f(n)u(n-i)}{N\sigma_x^2} \qquad (15)$$

Where, $\sigma_x = \frac{1}{N}\sum_{i=0}^{N-1} u^2(n-i)$

N is tap number of adaptive filter.
Step size is now bounded in the range of 0 to 2. It makes the convergence rate independent of signal powerd by normalizing the input vector with the energy of the input signal in the adaptive filter.

7 Simulation and Results

In order to evaluate the perforamnce and stability of the proposed method, we have simulated for a system that it requried wirelss Tele-Control system. The plant is chosen to be a Drone. Here, we are going to stablize the attitude of Drone by a conventional PID contoller and reduce theeffect of multiplath channel by equlizer. Attidude control consist of roll, pitch and yaw. The plant is MIMO 6 dimentional plant matrix with 3 inputs and outputs. As we disscused, in the case of Wirless Tele-Control System we have multipath channel certainly. In this situation we can consider that in the closed-loop system we would have two idenditical multipath channel, one for feedforward and the other for the feedback. In the result, in order to see the charachtresitc and influences in control system we show the singular values of open loop and the closed-loop system without equalizer. Thereafter, we simulate the step response of the closed-loop system with equzlier to confirm the stability and feasibility of equalizer which is implemented in closed-loopsystem. Following shows the Conditions of Simulation.

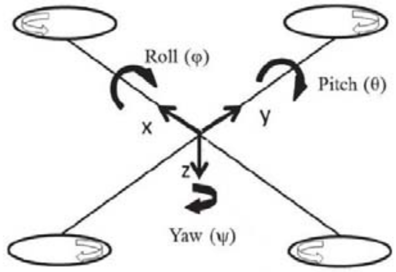

Fig.8 Attitude control of Drone

・Doyle Expression of the simplified atitude dynamics of Drone:

$$P(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$$

$$= \left[\begin{array}{cccccc|ccc}
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 5.1282 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 7.4074 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 7.4074 \\
\hline
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0
\end{array}\right]$$

where A , B, C and D are plant, input , output and direct matrices , respectivley .

・Controller of attitude:

$$K(s) = K_P\left(1 + \frac{1}{T_i s} + \frac{T_d s}{as+1}\right) I_{3\times 3}$$

$$= 0.0005\left(10 + \frac{0.01}{s} + \frac{2s}{0.01s+1}\right) I_{3\times 3}$$

・Channel Specification (with 10 taps M=10) :

$$H(s) = \sum_{i=1}^{M} \alpha_i e^{-sL_i} .$$

Feedforward channel and feedback channel are assumed to be identical.

Where, attenuated and time delay factor are indicated below.

$\alpha_i = rand(i)e^{\frac{-0.1i}{M}}$, $L_i = 0.5i \times sort(|rand(i)|)$

That rand(.) is uniformly distributed random numbers.

Results

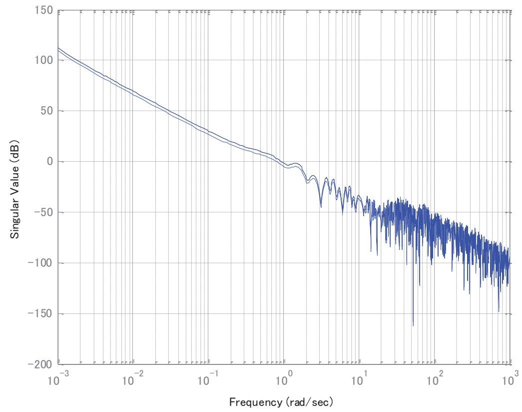

Fig.9 SVD of open loop without equalizer

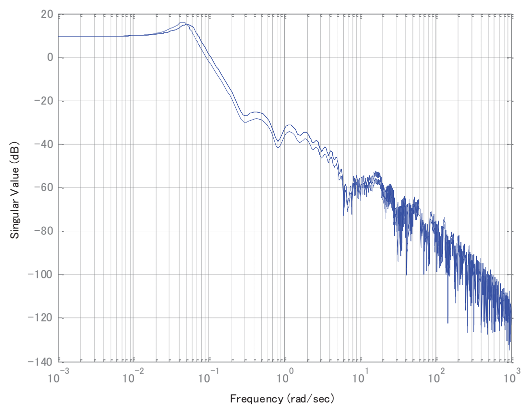

Fig. 10 SVD of closed loop without equalizer

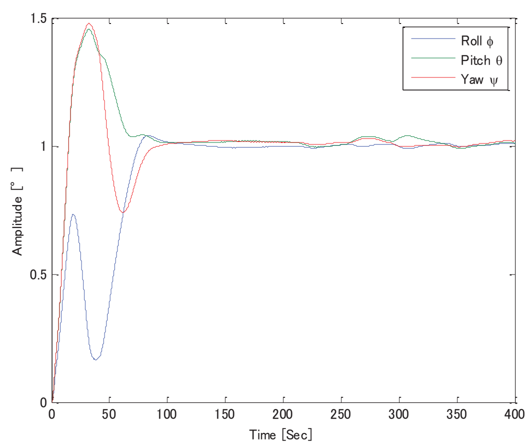

Fig. 11 Step response of closed-loop with equalizer

Fig. 9 shows the singular values of open loop that contains multipath channel without equalizer. As it is clear singular values in haigh frequencies, it flucuates frequently. Therefore, these kinds of charactrestic for open loop is not desired since it may casuse the intability of the closed-loop system. In Fig.10 we can see the singular values of the closed- loop system without equalizer. In this figure as well as Fig.9 in the high frequencies singular values fluctuates and in low frequencies it contain high gain that is not satisfying the stability condition. Thus, it would become unstable and sensitive to disturbances. As it is shown in Fig.11 eventhough roll and pitch angle have 50% overshoot, the step response of the closed-loop with equailzer is stable. Thus we could confirm the stablity of closed loop system even multipath channel exist in system. However, in this study we were not able to analyze the singular values of the closed loop and open loop with equilizer since equalizer is a time variant system . Therefore as a future work we are going to obtain the singular values with equalizer. In addtion, in this paper singular values of open loop and the closed-loop with equalizer could not be obtained due to equalizer. Since equalizer is a variant system singular values cant not be determined. Therefore, we also going obtain the singular values and even stability margin of system with equalizer in order to anaylize in frequency domain.

8 Conclusion

In this paper we implemented equalizer in Wireless Tele-Control system in order to get rid of instability in the closed-loop system which cuased by multipath channel. As a result we could confrim the stability of the closed-loop system with step response. However, the performance is not accepatble due to existence of overshoots. Moreover, in the actual case parameters of plant may be perturbed due to environmental and phisycal conditions. Therfore as a future work enhancment of performance and internal stability of MIMO system including parametres perturbation should be considered.

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